treatment, factor & levels

8. A grower wishes to compare three types of fertilizers as they affect crop yield.

9. A researcher wishes to investigate the effect of daily doses of vitamin C at doses of 200, 500, and 1000 mg on preventing the common cold.

8. level : 3 types of fertilizer
factor : fertilizer
treatment : 3 types of fertilizer

9. level : 200, 500, 1000 mg of vitamin C
factor : vitamin C
treatment : vitamin C

14. The proportion of defective items among 100 items selected from daily output is recorded for 25 days.

14. no, proportion of defective items are random

DATA SET DS1103 **8. Reducing Hostility** A psychologist wants to compare three methods for reducing hostility levels in university students. Eleven students were judged to have "great hostility," based on a certain psychological test (HLT). These students were then randomly divided into three groups—five were treated by method A, three were treated by method B, and the other three students were treated by method C. All treatments continued throughout a semester, at the end of which the HLT test was given again. The results are shown in the table.

Method	Scores on the HLT Test				
A	73	83	76	68	80
B	54	74	71		
C	79	95	87		

a. Perform an analysis of variance for this experiment.

b. Do the data provide sufficient evidence to indicate a difference in mean student scores after treatment for the three methods?

a.

source	df	SS	MS	F
treatment	2	641.88	$\frac{SST}{k-1} = 320.94$	$\frac{MST}{MSE} = \frac{320.94}{59.83} = 5.36$
error	8	478.67	$\frac{478.67}{8} = 59.83$	
Total	10	1120.55		

$F_{stat} = \frac{MST}{MSE} \sim F_{(k-1),(n-k)}$

$\bar{A} = 76$ $\quad \sum x_i^2 = 28938$

$\bar{B} = 66.33$ $\quad \sum x_i^2 = 13433$

$\bar{C} = 87$ $\quad \sum x_i^2 = 22835$

$\bar{G} = \frac{840}{11} = 76.3637$

total SS $= \sum (x_{ij} - \bar{G})^2 = 1120.55$

$SST = \sum_{i=1} n_i (\bar{T}_i - \bar{G})^2 = 641.88$

SSE = total SS − SST = 1120.55 − 641.88 = 478.67

RR

$F_{2,8,0.05} = 4.46$

since $F^* = 5.36$ is within RR, reject H_0

DATA SET DS1104

10. Assembling Electronic Equipment An experiment was conducted to compare the effectiveness of three training programs, A, B, and C, for assemblers of a piece of electronic equipment. Five employees were randomly assigned to each of three programs. After completion of the program, each person assembled four pieces of the equipment, and their average assembly time was recorded. Several employees resigned during the course of the program; the remainder were evaluated, producing the data shown in the accompanying table. Use the *Excel* printout to answer the questions in parts a–d.

Training Program	Average Assembly Time (min)				
A	59	64	57	62	
B	52	58	54		
C	58	65	71	63	64

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
A	4	242	60.5	9.667
B	3	164	54.667	9.333
C	5	321	64.2	21.7

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	170.45	2	85.225	5.704	0.0251	4.256
Within Groups	134.467	9	14.941			
Total	304.917	11				

a. Do the data indicate a significant difference in mean assembly times for people trained by the three programs? Give the p-value for the test and interpret its value.

b. Find a 99% confidence interval for the difference in mean assembly times between persons trained by programs A and B.

c. Find a 99% confidence interval for the mean assembly times for persons trained by program A.

d. Do you think the data will satisfy (approximately) the assumption that they have been selected from normal populations? Why?

a) ① $H_0: \mu_A = \mu_B = \mu_C$ H_1: there is a diff b/w 3

② $\alpha = 0.05$

③ $F_{STAT} = \frac{MST}{MSE} \sim F_{(k-1),(n-k)}$

source	SS	df	MS	F
treatment	170.45	2	$\frac{SST}{df} = \frac{170.45}{2} = 85.225$	$F = \frac{MST}{MSE} = \frac{85.225}{14.941} = 5.704$
Error	134.467	9	$\frac{SSE}{n-k} = \frac{134.467}{9} = 14.941$	
total	304.917	11		

$\bar{A} = 60.5$ $G = 727$

$\bar{B} = 54.667$ $\bar{G} = 60.5833$

$\bar{C} = 64.2$

total SS: $\sum (X_{ij} - \bar{G})^2 = 304.917$

$SST = \sum n_i (\bar{T}_i - \bar{G})^2 = 170.25$

SSE = total SS − SST = 134.467

$P(F > 5.704) = 0.0251$

∴ since p-value of $F^* < 0.05$, we reject H_0.

b) 99% CI: $(\bar{A} - \bar{B}) \pm t_{n;\frac{\alpha}{2}} \sqrt{MSE\left(\frac{1}{n_A} + \frac{1}{n_B}\right)}$

$= 60.5 - 54.667 \pm 3.25 \sqrt{14.941\left(\frac{1}{4} + \frac{1}{3}\right)}$

$= (-3.767, 15.433)$

since 0 lies within the range, the means are similar

c) 99% CI: $\bar{A} \pm t_{n;\frac{\alpha}{2}} \sqrt{\frac{MSE}{n_A}} = 60.5 \pm 3.25 \sqrt{\frac{14.941}{4}}$

$= (54.22, 66.28)$

d) yes, assembly times are sums of many small independent tasks and these normally follow a normal distribution.

DATA SET DS1108

14. Tuna Fish The estimated average prices for a variety of brands of tuna fish, based on prices paid nationally, are shown here.[1]

Light Tuna in Water		White Tuna in Water	White Tuna in Oil	Light Tuna in Oil	
.99	.53	1.49	1.27	2.56	.62
1.92	1.41	1.29	1.22	1.92	.66
1.23	1.12	1.27	1.19	1.30	.62
.85	.63	1.35	1.22	1.79	.65
.65	.67	1.29		1.23	.60
.69	.60	1.00			.67
.60	.66	1.27			
		1.28			

Source: From "Pricing of Tuna" Copyright 2001 by Consumers Union of U.S., Inc., Yonkers, NY 10703-1057, a nonprofit organization. Reprinted from the June 2001 issue of *Consumer Reports*® for educational purposes only. No commercial use or reproduction permitted. www.ConsumerReports.org®.

a. Construct the ANOVA table for a completely randomized design.

b. Is there evidence of a significant difference in average price for these packages at the $\alpha = .05$ level of significance? At the $\alpha = .01$ level of significance?

c. Find a 95% confidence interval estimate of the difference in price between light tuna in water and light tuna in oil.

d. Find a 95% confidence interval estimate of the difference in price between white tuna in water and white tuna in oil.

e. What other confidence intervals might be of interest to the researcher who conducted this experiment?

a) $\bar{A} = 0.88$ $\quad n_A = 14$

$\bar{B} = 1.28$ $\quad n_B = 8$

$\bar{C} = 1.225$ $\quad n_C = 4$

$\bar{D} = 1.242$ $\quad n_D = 6$

$\bar{G} = \frac{34.91}{32} = 1.091$

total SS: $\sum(x_{ij} - \bar{G}) = 8.27$

$SST = \sum_{i=1} n_i (\bar{T} - \bar{G})^2 = 1.117$

$SSE = \text{total SS} - SST$

$= 8.21 - 1.117$

$= 7.093$

source	SS	df	MS	F^*
treatment	1.117	3	$MST = \frac{SST}{k-1} = 0.3723$	$F = \frac{MST}{MSE} = \frac{0.3723}{0.2533} = 1.47$
error	7.093	28	$MSE = \frac{SSE}{n-k} = 0.2533$	
total	8.21	31		

b) $F_{3,28;0.05} = 2.95$

$F_{3,28;0.01} = 4.57$

since the F^* $< F_{3,28;0.05}$ & $< F_{1;28;0.01}$, we fail to reject the

c) $95\%\ CI = (\bar{A} - \bar{D}) \pm t_{28;0.025}\sqrt{MSE\left(\frac{1}{n_A} + \frac{1}{n_D}\right)}$

$= (0.88, 1.242) \pm 2.048\sqrt{0.2533\left(\frac{1}{14} + \frac{1}{6}\right)}$

$= -0.362 \pm 0.502$

$= (-0.864, 0.140)$

since 0 lies within the range, the means are the same

d) $95\%\ CI = (\bar{B} - \bar{C}) \pm t_{28;0.025}\sqrt{MSE\left(\frac{1}{n_B} + \frac{1}{n_C}\right)}$

$= (1.28 - 1.225) \pm 2.048\sqrt{0.2533\left(\frac{1}{8} + \frac{1}{4}\right)}$

$= 0.055 \pm 0.631$

$= (-0.576, 0.686)$

since 0 lies within the range, the means are the same.

e) either water vs oil or white vs light.